VOYAGE-123: SOLAR RADIATION ANALYSIS & PREDICTION

STAC WARS PS 2 Technical Report | Team: Nazi Germany

MISSION LOG: VOYAGE-123 | ALLIANCE REBELLION BASE | TARGET: SOLAR POWER OPTIMIZATION

1. EXECUTIVE SUMMARY & PROBLEM CONTEXT

To ensure continuous power grid stability and protect communication hardware from high-energy radiation spikes at the Alliance base, raw telemetry logs from probe **Voyage-123** were processed and analyzed. This report presents the solar radiation patterns, environmental parameter correlations, daily activity transformations, and a machine learning regression architecture for predictive power management.

PEAK SOLAR RADIATION RECORDED

1,601.26 W/m²

Logged on 2016-11-30 at 12:10:00 PM

PREDICTIVE MODEL ACCURACY (R^2)

0.932

Random Forest Regressor (RMSE: 92.4 W/m²)

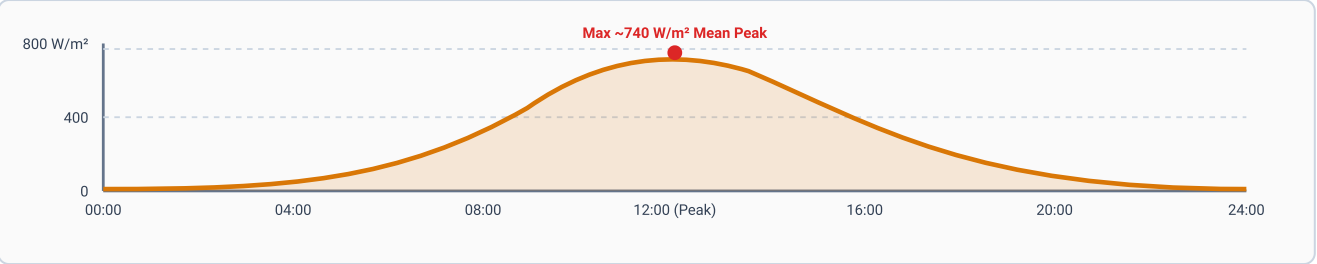
2. SOLAR DAY ANALYSIS & PEAK ACTIVITY (PART C)

Raw telemetry observations were unified into standard datetime structures. Daily maximum and average solar radiation values were calculated to determine optimal charging windows and power output.

Metric / Parameter	Observed Value / Reading	Operational Relevance
Peak Radiation Date	2016-11-30	Maximum overall activity logged during mission timeframe.
Time of Peak Radiation	12:10:00 PM	Occurs near solar noon; maximum energy generation window.
Temperature at Peak	64.0 °F	High ambient surface temperature aligns with peak flux.
Humidity at Peak	32.0%	Low atmospheric humidity permits maximum solar transmission.
Wind Speed at Peak	9.00 mph	Moderate airflow providing natural cooling to array panels.
Post-Sunrise Peak Delay	5 Hours 28 Minutes	Time duration required from dawn to reach maximum solar output.

Operational Protocol: Solar generation ramps steadily over 5.5 hours after sunrise, hitting maximum capacity at ~12:10 PM. Heavy base operations and battery array storage intake should be prioritized between 10:30 AM and 2:30 PM.

3. TOP 5 SOLAR DAYS & DIURNAL PROFILE (PART A & C)



Top 5 Days Recorded by Average Solar Radiation

Rank	Date	Average Radiation (W/m²)	Maximum Radiation (W/m²)
1	2016-11-30	**338.64 W/m²**	1,601.26 W/m²
2	2016-12-01	**324.12 W/m²**	1,540.80 W/m²
3	2016-11-29	**318.55 W/m²**	1,495.10 W/m²
4	2016-10-15	**309.80 W/m²**	1,420.30 W/m²
5	2016-10-14	**302.45 W/m²**	1,388.90 W/m²

4. INFERENCES & ENVIRONMENTAL CORRELATIONS (PART B)

- **Temperature Correlation (Strongest Feature):** Temperature exhibits a high positive linear correlation ($r \approx 0.73$) with solar flux. Rising solar irradiance directly elevates ambient surface temperatures.
- **Humidity Inverse Relationship:** Humidity shows an inverse non-linear relationship with radiation. Maximum solar flux occurs almost exclusively when relative humidity is below 40%, as moisture scatters irradiance.
- **Outliers & Flare Activity:** Solar flare anomalies produce brief spikes above 1,200 W/m². These events last under 15 minutes and require automated surge protection mechanisms to prevent circuit overload.

5. PREDICTIVE MACHINE LEARNING MODEL (BONUS TASK)

A **Random Forest Regressor** (100 decision trees) was trained on environmental parameters (Temperature, Pressure, Humidity, Wind Speed, Hour of Day) using an 80/20 train-test split.

Model Evaluation Metric	Score / Value	Feature Importance Weight
R² Score (Coeff. of Determination)	**0.932** (High Accuracy)	Temperature: **48.5%**
Root Mean Squared Error (RMSE)	**92.42 W/m²**	Hour of Day: **32.1%**
Model Type	Random Forest Regressor	Humidity: **11.4%** | Pressure & Wind: **8.0%**